

Quantum Benchmarks

Hardware Runs

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This benchmarking report was generated using Classiq v1.10.1.

The report includes hardware runs for a collection of pre-defined and user-defined functions and algorithms, where each benchmark can be instantiated at different problem sizes. For each benchmark experiment, we report a score defined specifically for that example, together with the execution time¹ and quantum-program properties obtained through Classiq's hardware-aware synthesis². For further information, see the Benchmarking directory in classiq-library.

¹Currently, the reported time includes both execution time and waiting time.

²The properties of the actual circuit executed on the hardware may differ slightly due to pre-run transpilation.

GHZ State Preparation

We benchmark the preparation of the n -qubit GHZ state,

$$\frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}.$$

Following the population-and-parity fidelity estimator of Monz et al., PRL 106, 130506 (2011), the score combines the population $P = p_0 + p_{2^n-1}$ with the coherence C , extracted from the amplitude of the parity oscillation $\Pi(\phi) = P_{\text{even}}(\phi) - P_{\text{odd}}(\phi)$ under a collective analysis rotation. The resulting fidelity estimator is

$$F = \frac{P + C}{2}.$$

For more details see Benchmark GHZ notebook.

ghz - 3 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	4	2
Classiq	simulator_statevector	1.0	0.0	4	2